

Wear OS

Interactive Systems Design

Laurea Magistrale in Informatica

Anno Accademico 2018-2019



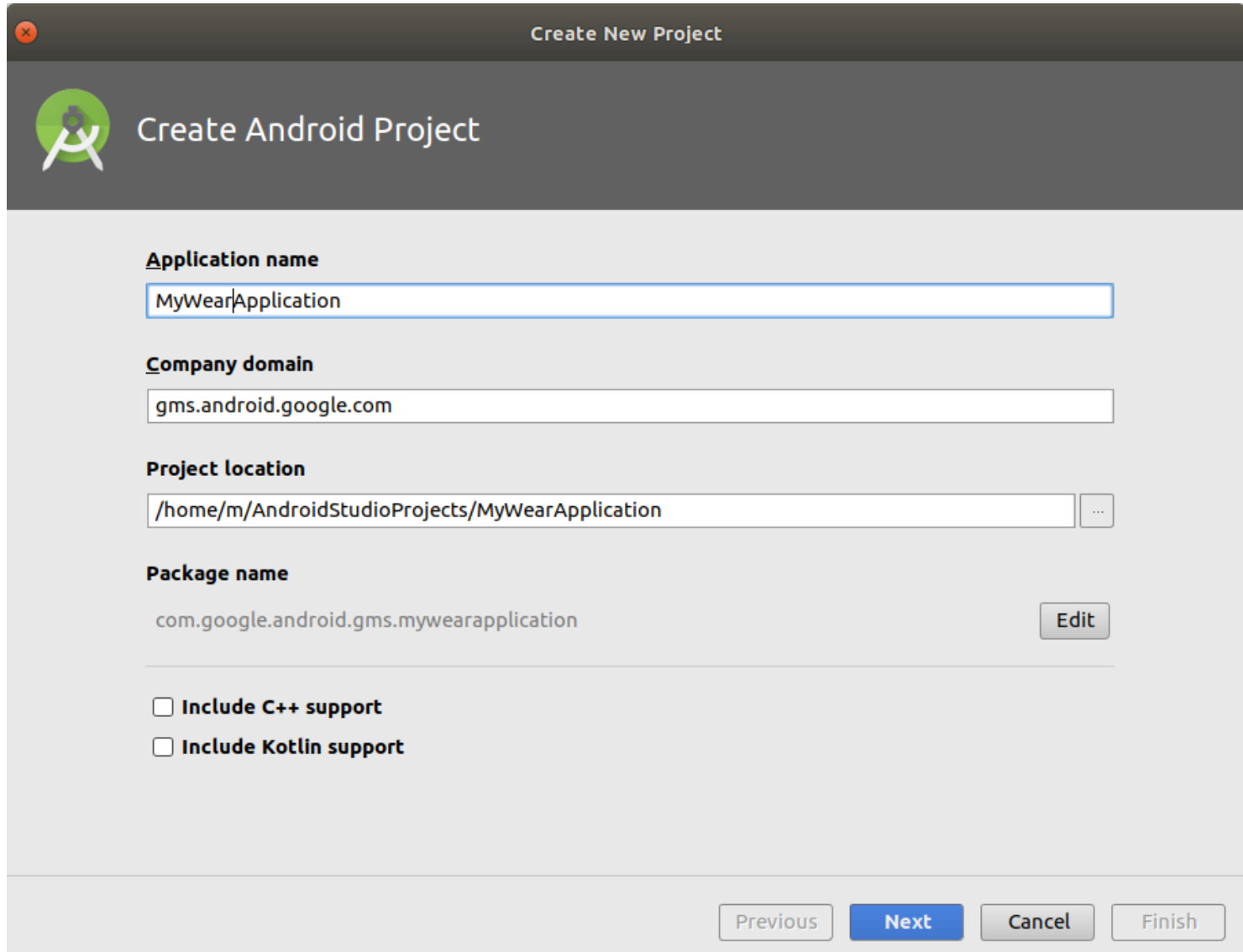
Wear OS

- ❑ Conosciuto precedentemente come **Android Wear**
- ❑ Versione di Android progettata per smartwatch ed altri dispositivi indossabili
 - Un dispositivo Wear OS deve essere associato ad uno smartphone
 - And es. per condividere notifiche, installare app, ecc.
- ❑ Dispositivi disponibili (2018):
 - Smartwatch con schermo circolare
 - Smartwatch con schermo quadrato (sempre più rari)

- API basata su quella di Android
 - Aggiunge tipi diversi di widget dedicati
 - Aggiunge funzionalità per le notifiche
 - A partire dalla versione Android Wear 2.0 sono disponibili metodi di immissione del testo
 - Un basato softkeyboard (con gesti)
 - Uno su handwriting
 - And es. per condividere notifiche, installare app, ecc.

Android Studio

Creazione nuovo progetto (1)



The screenshot shows the 'Create New Project' dialog in Android Studio. The dialog has a title bar with a close button and the text 'Create New Project'. Below the title bar is a header section with the Android logo and the text 'Create Android Project'. The main area contains several input fields and checkboxes. The 'Application name' field is filled with 'MyWearApplication'. The 'Company domain' field is filled with 'gms.android.google.com'. The 'Project location' field is filled with '/home/m/AndroidStudioProjects/MyWearApplication' and has a browse button (three dots). The 'Package name' field is filled with 'com.google.android.gms.mywearapplication' and has an 'Edit' button. At the bottom, there are two checkboxes: 'Include C++ support' and 'Include Kotlin support', both of which are unchecked. At the very bottom of the dialog are four buttons: 'Previous', 'Next', 'Cancel', and 'Finish'.

Create New Project

Create Android Project

Application name
MyWearApplication

Company domain
gms.android.google.com

Project location
/home/m/AndroidStudioProjects/MyWearApplication ...

Package name
com.google.android.gms.mywearapplication Edit

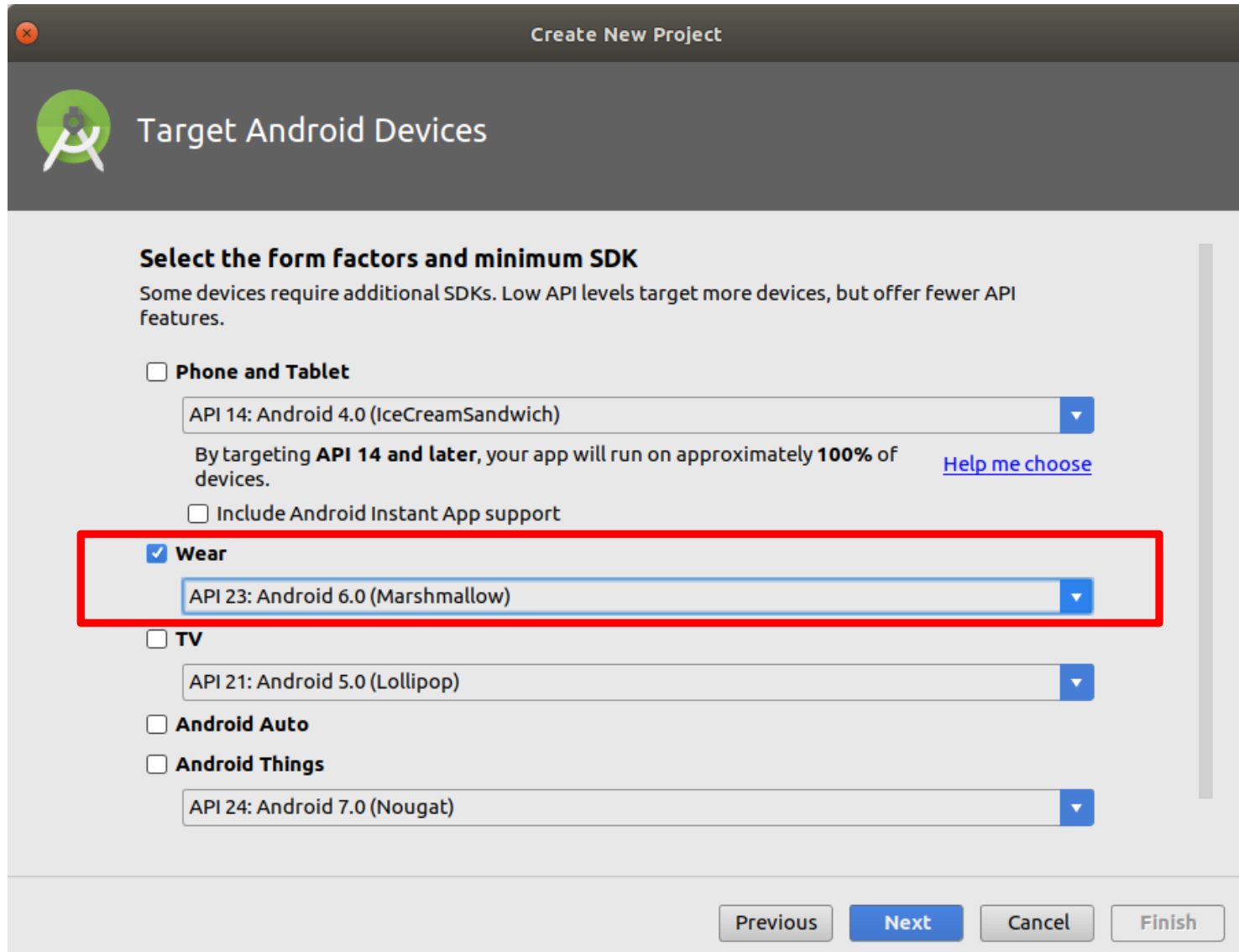
☐ Include C++ support

☐ Include Kotlin support

Previous Next Cancel Finish

Android Studio

Creazione nuovo progetto (2)



Create New Project

 **Target Android Devices**

Select the form factors and minimum SDK
Some devices require additional SDKs. Low API levels target more devices, but offer fewer API features.

☐ **Phone and Tablet**
API 14: Android 4.0 (IceCreamSandwich) ▼
By targeting **API 14 and later**, your app will run on approximately **100%** of devices. [Help me choose](#)
☐ Include Android Instant App support

☒ **Wear**
API 23: Android 6.0 (Marshmallow) ▼

☐ **TV**
API 21: Android 5.0 (Lollipop) ▼

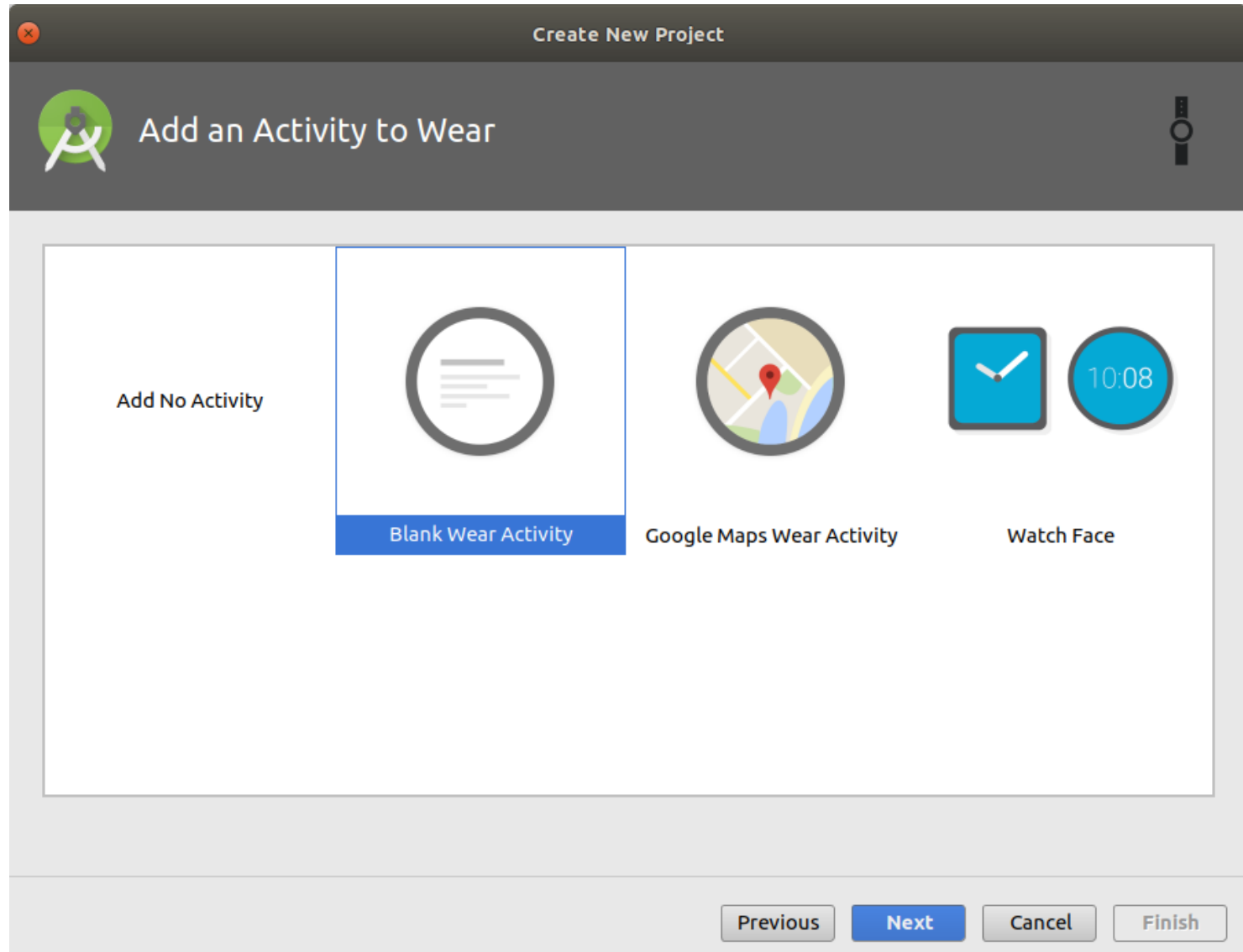
☐ **Android Auto**

☐ **Android Things**
API 24: Android 7.0 (Nougat) ▼

Navigation: Previous Next Cancel Finish

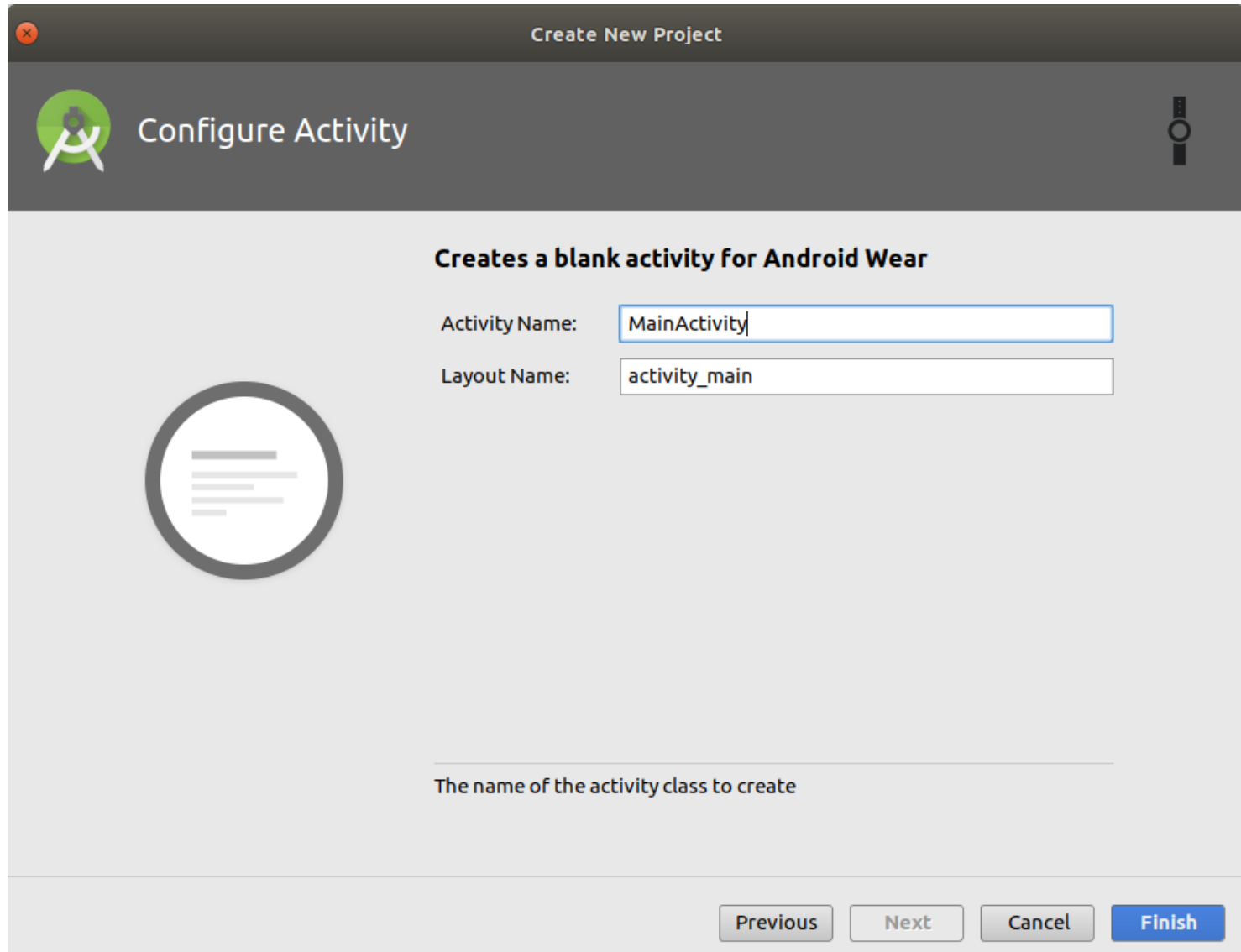
Android Studio

Creazione nuovo progetto (3)



Android Studio

Creazione nuovo progetto (4)



Create New Project

 Configure Activity 

Creates a blank activity for Android Wear

Activity Name:

Layout Name:



The name of the activity class to create

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.google.android.gms.mywearapplication">

    <uses-feature android:name="android.hardware.type.watch" />

    <uses-permission android:name="android.permission.WAKE_LOCK" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="MyWearApplication"
        android:supportsRtl="true"
        android:theme="@android:style/Theme.DeviceDefault">
        <uses-library
            android:name="com.google.android.wearable"
            android:required="true" />

        <!--
            Set to true if your app is Standalone, that is, it does not require the handheld
            app to run.
        -->
        <meta-data
            android:name="com.google.android.wearable.standalone"
            android:value="true" />

        <activity
            android:name=".MainActivity"
            android:label="MyWearApplication">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```




MainActivity.java

```
package com.google.android.gms.mywearapplication;
import ...
public class MainActivity extends WearableActivity {
    private TextView mTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        mTextView = (TextView) findViewById(R.id.text);

        // Enables Always-on
        setAmbientEnabled();
    }
}
```



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.wear.widget.BoxInsetLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/dark_grey"
    android:padding="@dimen/box_inset_layout_padding"
    tools:context=".MainActivity"
    tools:deviceIds="wear">

    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:padding="@dimen/inner_frame_layout_padding"
        app:boxedEdges="all">

        <TextView
            android:id="@+id/text"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/hello_world" />

    </FrameLayout>
</android.support.wear.widget.BoxInsetLayout>
```



Porting di un'app per smartphone

- ❑ Aggiungere a `AdroidManifest.xml` `<uses-feature android:name="android.hardware.type.watch" />`
- ❑ È comunque possibile continuare ad estendere **Activity** invece di **WearableActivity**
- ❑ Non è indispensabile utilizzare i widget dedicati (ad es. `android.support.wear.widget.BoxInsetLayout`)
- ❑ Il passaggio inverso (smarwatch → smartphone) è più complicato (la libreria `com.google.android.wearable` non è presente su smartphone)



BubbleBoard_v1

- ❑ Prima versione della tastiera “a bolle”
- ❑ Porting di un'app per smartphone
 - Eseguitibile su smartphone se si rimuove `<uses-feature android:name="android.hardware.type.watch" />` da `AndroidManifest.xml`
- ❑ Caricata su GitLab
 - https://gitlab.com/isd18-19/projectStub/BubbleBoard_v1.git
- ❑ Oltre alle funzionalità tipiche di una tastiera, l'app:
 - mostra all'utente la frase da trascrivere
 - memorizza in file di log le azioni dell'utente
 - calcola metriche (velocità, accuratezza) e le salva in file di log
- ❑ Richiede permesso *Storage*



Classi

- ❑ `TextEntryMetrics`: calcolo delle metriche
- ❑ `TriLayoutPredictor`, `TriSLayoutPredictor`: servono a predire il layout migliore in base agli ultimi tre caratteri
 - Precomputati utilizzando il Leipzig Corpora News-typical 2016
 - Andranno ricreati per tipi di layout diversi
- ❑ `Keyboard`: creazione/modifica layout della tastiera
- ❑ `strings.xml` contiene le frasi del dataset di MacKenzie <http://www.yorku.ca/mack/phrases2.txt>